INCH-POUND

A-A-54717
10 September 1992

COMMERCIAL ITEM DESCRIPTION

COMPRESSOR AND OXYGEN BLENDER, VENTILATOR

(Portable, Battery and Line Powered)

The General Services Administration has authorized the use of this Commercial Item Description.

This Commercial Item Description covers a compressor with oxygen blender, AC power supply, and other required components.

Salient Characteristics:

The portable compressor system shall be suitable for use in a hospital, ambulance, military aeromedical aircraft, and the field where no external power is available. It shall provide dry, 100 percent oil-free, compressed air and oxygen blended mixtures. The system shall be compatible with a specific volume ventilator specified elsewhere in the procurement document.

A mechanical means shall be provided to attach this compressor system to the ventilator. Such mechanical means shall be compatible with the ventilator.

The complete portable compressor system shall include the following: a self contained portable compressor, integral or separate air/oxygen blender, integral or separate AC power supply unit, accumulator type compressed air storage reservoir, battery recharge circuitry, rechargeable battery, voltage selector switch, low battery alarm, power connection indicator, electrical connections for external power sources, gas filter, pneumatic connections, condensed operating instructions, dedicated compressor case with integral carrying handle(s), maintenance and operating instructions, and the additional components required to form an operational ventilation system when combined with the ventilator. The condensed operating instructions shall be marked on this

AMSC N/A FSC 6530

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dedicated case or on a label attached to this case. Controls and indicators shall be functionally grouped to facilitate operator usage. The complete compressor system shall be contained in a carry-all carrying case with handle.

The compressor shall be capable of sustaining a positive pressure of approximately 7 psig and a peak airflow of at least 40 Lpm. It shall be designed with compressor unit components that interconnect with the ventilator to allow the ventilator to operate using 7 psig air and oxygen blended input. All ventilator functions shall be operational and all ventilator controls shall be either directly operational or superseded by corresponding controls on the compressor system.

The compressor shall be capable of allowing the ventilator to deliver the following peak tidal volume:

a. 0 to 300 ml in 0.5 seconds
b. 0 to 700 ml in 1.0 seconds
c. 0 to 900 ml in 1.5 seconds
d. 0 to 1100 ml in 2.0 seconds

The compressor shall have a minimum operating temperature range of minus 4°F to plus 115°F from all power sources cited below, including the internal rechargeable battery. The minimum long term storage temperature range shall be from 50°F to 80°F.

The compressor shall contain both audible and visual low battery alarms which activate when 10 minutes (at least) of battery power remains. The audible alarm shall be mutable.

The compressor shall be provided with an indicator which denotes the presence and source (internal/external) of operating power.

A mechanical means shall be provided to allow attachment of the compressor, with blender, to the ventilator to form a composite ventilator system.

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The compressor with the AC power supply unit shall be operational from any of the following sources:

a. Internal battery.

b. Nominal 12 volts DC, vehicle generating system rated at 120 amperes at 14 volts DC with negative ground polarity.

c. External 120 amp-hour heavy duty 12 volt battery with negative ground polarity.

d. Aircraft 28 volt DC generating system with negative ground polarity.

e. 115/230 volt, 50-400 Hz.

The self-contained portable compressor shall be housed in a semi-rigid plastic or corrosion-resistant metal case with a carrying handle. This case shall house only the self-contained portable compressor (and integral blender and/or integral AC power supply unit if they are not adjunctive). It and all other components, including operating and maintenance manuals, shall be contained in a carry-all carrying case with handle(s).

The circuitry of the self-contained compressor shall contain electrical connections to allow both battery recharge and compressor operation from all the electrical sources listed herein.

The internal, rechargeable battery shall be 12 volts (2 batteries wired together to provide 12 volts is acceptable). This battery shall have the capability to allow the compressor to deliver compressed gas to the ventilator, allowing the ventilator to be operational for at least one hour without diminished ventilator capability (peak flow 40 Lpm with inspiratory time set at one second).

This internal battery shall not deplete more than approximately 1.5 percent per month, or 18 percent per year if stored at 70°F. It shall be rechargeable, complete charge within 24 hours, from any of the following sources (with the AC power supply unit):

a. 115/230 volt, 50-400 Hz.

b. Nominal 12 volt DC, vehicle generating system rated at 120 amperes at 14 volts DC with negative ground polarity.

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d. Aircraft 28 volt DC generating system with negative ground polarity.

The gas blender shall either be integral or adjunctive to the compressor. The blending mechanism shall be capable of proportionately mixing medical grade oxygen with compressed air. Blender settings shall permit delivery of oxygen concentrations ranging between 21 percent and 100 percent.

The AC power supply shall be either integral or adjunctive to the portable compressor apparatus. This supply shall operate from nominal 115/230 V, 50-400 Hz, single phase power sources and shall contain a voltage selector switch. This AC power supply shall contain an appropriately strain-relieved line cord, at least 6 feet long, consisting of SJT-type wire with a minimum rating of 18/3 AWG, and terminated in a 3-conductor male hospital-grade plug. If the AC power supply is integral to the compressor, a means by which to wrap and secure the AC power cord shall be provided. If the AC power supply is adjunctive, its output cable shall be at least 4 feet long and terminate in a suitable, non-shorting type plug to interface with the compressor. The power supply output voltage shall be within the range of acceptance of the compressor apparatus described herein, for both battery recharge and operation. Ground polarity shall be negative.

Connecting hose(s) and pneumatic connections shall be provided with the portable compressor. If the blender is integral to the compressor, one connecting hose for attachment to the ventilator and one hose for a remote 50 psig oxygen source shall be provided. If the blender is adjunctive to the compressor, three hoses shall be provided: one hose shall connect the compressor output to the blender air input; one hose shall connect a remote oxygen source to the blender oxygen input; and one hose shall connect the blender output to the gas input of the ventilator. All hoses shall meet the following criteria:

a. Conform to CGA-9 standard for color-coded markings of medical gas apparatus.

b. Be rated for 200 psi working pressure.

c. Be odorless, tasteless and exhibit no toxicity.

d. Include appropriate DISS fittings with swivel attachment nuts to prevent intermixing of dissimilar gas.

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A vehicle power cord assembly, approximately 6 feet long, having a vehicle cigar lighter plug on one end and a plug to mate with the portable compressor on the other end shall be provided.

An aircraft power cord assembly, approximately 6 feet long shall be provided having one free end (pigtail) and the other end with a plug to mate with the compressor.

The complete compressor system shall be contained in a carry-all carrying case with handle. This case shall be in addition to the dedicated case for the self-contained compressor.

The self-contained compressor, less all other components, shall not weigh more than 20 pounds. If the compressor includes an integral blender and AC power supply, its combined weight shall not exceed 30 pounds.

The self-contained compressor, less all other components, shall have a volume less than 2.0 cubic feet. If the compressor includes an integral blender and AC power supply, its combined volume shall not exceed 2.5 cubic feet.

The apparatus shall meet the requirements of UL 544/Medical and Dental Equipment.

Two complete sets of instructions shall be provided for operation, maintenance, and repair.

Workmanship. The ventilator compressor and oxygen blender shall be free from defects which detract from their appearance or impair their serviceability.

Unit. Each (EA). One compressor and oxygen blender ventilator, as specified, constitutes one unit.

Quality Assurance Provisions.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The

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Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105, with an AQL of 0 (percent defective) and an inspection level of 1. The unit of product for examination purposes shall be one ventilator compressor and oxygen blender.

Sampling for tests. Sampling for tests, including dimensional test, shall be conducted in accordance with MIL-STD-105, with an AQL of 0 (percent defective) and an inspection level of 1. The unit of product for test purposes shall be one ventilator compressor and oxygen blender.

Tests. Tests shall be conducted to determine compliance with specification requirements. Where feasible, the same sample shall be used for the determination of two or more test characteristics.

Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, providing they fall within the tolerances specified using conversion tables contained in the latest revision of Federal Standard 376, and all other requirements of this document are met.

If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable.

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The contracting officer has the option of accepting or rejecting the product.

Contractor certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this Commercial Item Description, and that the product conforms to the producers' own drawings, specifications, standards, and quality assurance practices. The government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Regulatory requirements.

Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging, and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

Ordering data. Intermediate/exterior package quantities, labeling, and marking must be specified in the contract and/or order.

NOTES: 1. The ventilator compressor and air/oxygen blender is intended to be compatible with a specific portable volume ventilator, generic NSN: 6530-01-292-1049.

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2. The requisitioner is required to identify the ventilator by brand name and part number when ordering this compressor.

3. The brand name and part number of the ventilator, that this compressor is to be compatible with, must be specified in the procurement document.

4. The following National Stock Number is covered by this document.

6530-01-319-8234

MILITARY INTERESTS:

Custodians:

Air Force - 03
Army - MD
Navy - MS

PREPARING ACTIVITY:

DoD-MB

Agent:

DLA-DM

CIVIL AGENCY COORDINATING ACTIVITIES:

FDA-MPQAS
USPHS
VA - OSS

Project Number: 6530-2357

Location: CMPRS12C.DOC

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

1. RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	A-A-54717	a0 September 1992

3. DOCUMENT TITLE: COMPRESSOR AND OXYGEN BLENDER, VENTILATOR (Portable, Battery and Line Powered)

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	e. DATE SUBMITTED (YYMMDD)

7. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
DEFENSE PERSONNEL SUPPORT CENTER ATTN: DPSC-MSE	(1) Commercial AC 215-737-2870	(2) AUTOVON AV 444-2870
c. ADDRESS (Include Zip Code) 2800 SO. 20TH STREET PHILADELPHIA, PA 19145	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

DD Form 1426, OCT 89 — Previous editions are obsolete — 98/290

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